JOTO MJINI

On hot mornings, central Nairobi's ground is 12.8 °C hotter than its forests

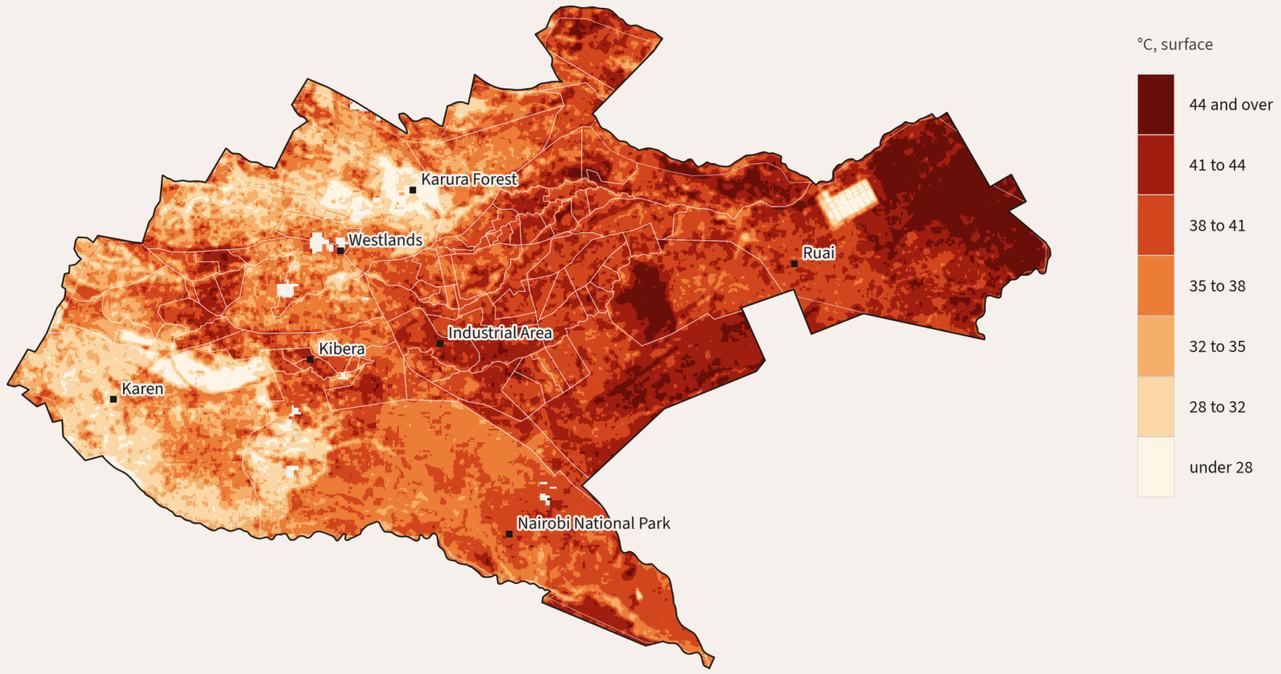
Mapping urban heat and cooling inequality across Nairobi with open satellite data

Stephane [surname], data analyst, Nairobi
Land, Food and Heat Resilience in Africa's Arid Regions, Nairobi, 26 November 2026

The ground in the centre was 12.8 °C hotter than under Karura's trees

On hot mornings, Nairobi's east and centre are hottest at the surface

Land surface temperature, median of 12 clear Landsat 8 and 9 mornings, January to March 2023 to 2026. County median 38.7 °C.



Source: USGS Landsat 8 and 9 Collection 2 via Microsoft Planetary Computer; analysis Joto Mjini. Ward lines: geoBoundaries (Omare and Omare), CC BY 4.0. Labels © OpenStreetMap contributors.

38.7 °C

half of the county's ground was hotter than this

40.5 °C

city centre and Industrial Area

27.6 °C

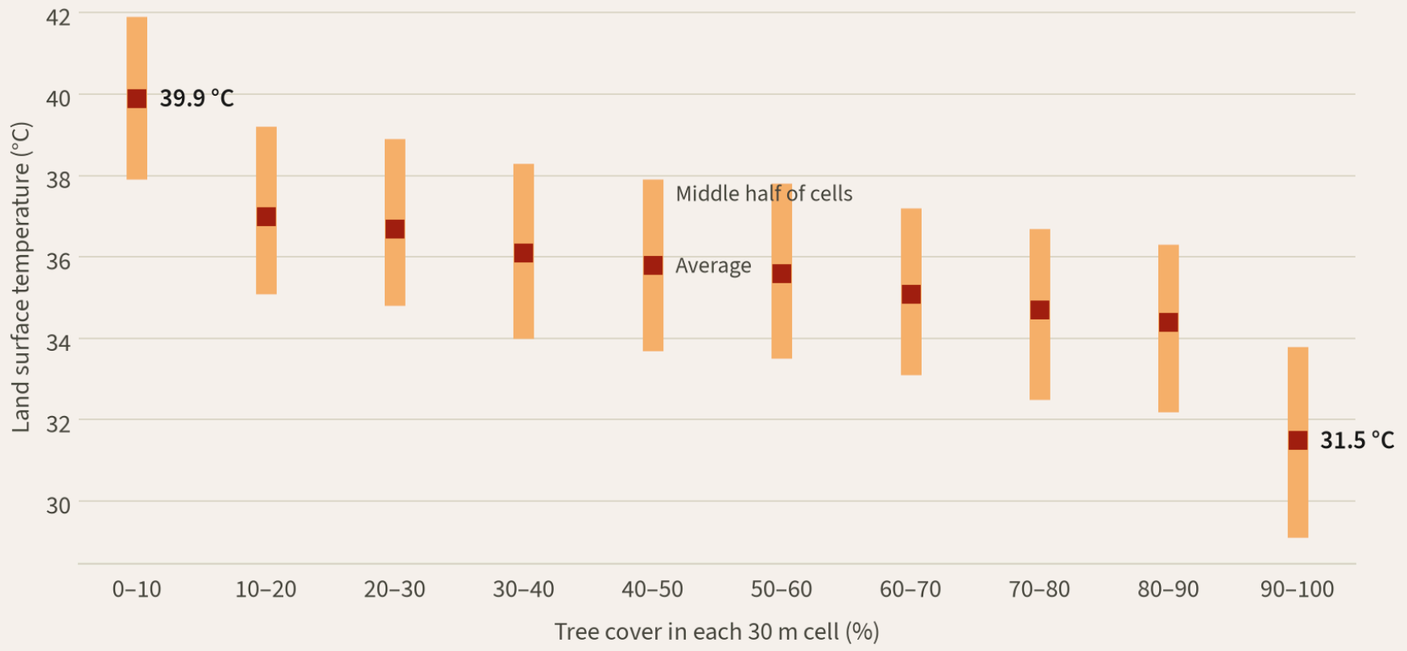
Karura Forest and the Arboretum

Land surface temperature, not air temperature. Clear mornings, January to March 2023 to 2026.

Where there are trees, the ground is cooler

The more tree cover, the cooler the ground

Average land surface temperature for each band of tree cover, Nairobi County, January to March mornings 2023 to 2026.



Source: USGS Landsat 8 and 9 Collection 2 via Microsoft Planetary Computer; analysis Joto Mjini. Tree cover: ESA WorldCover 2021, CC BY 4.0.

7.8 °C

hotter: mostly built-up ground against ground mostly under trees

39.9 °C

under 10 % tree cover

31.5 °C

over 90 % tree cover

An association. Dry grass and bare ground were as hot as built-up land in these months.

About 827,000 people live on the hottest fifth of the ground

The ten wards where people live on the hottest ground

Population-weighted land surface temperature, hot-season mornings 2023 to 2026.

Ward	Surface temp. (°C)	People	Tree cover
1. Viwandani	42.2	49,100	2 %
2. Kangemi	42.2	11,700	13 %
3. Kawangware	41.9	56,500	10 %
4. Dandora Area I	41.7	12,900	3 %
5. Babadogo	41.4	42,500	8 %
6. Kayole South	41.4	38,200	1 %
7. Kiamaiko	41.3	45,700	2 %
8. Kariobangi North	41.3	39,100	5 %
9. Dandora Area II	41.2	28,600	8 %
10. Upper Savannah	41.2	23,200	3 %

Sources: Landsat (USGS); WorldPop 2025 (CC BY 4.0); wards geoBoundaries (CC BY 4.0).

4.4 °C

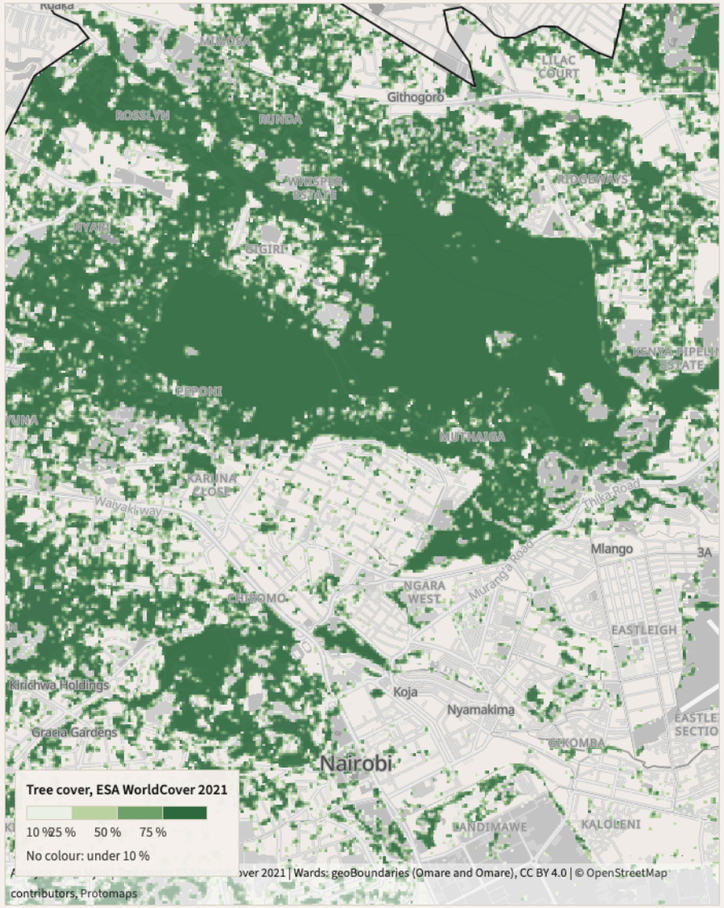
hotter ground for residents of the hottest fifth of wards than the coolest fifth

3.3 %

tree cover in the hottest fifth of wards, against **36.1 %**

This measures exposure: where people live on hot ground with little tree cover.

Islands of cool: Karura Forest and the Arboretum



5 of 9

Islands of cool

27.5 °C

Green on this layer is tree cover. The canopy of Karura Forest was 27.5 °C on hot mornings, and the Nairobi Arboretum 28.8 °C. Built-up land within 100 m of Karura was 2.1 °C cooler than built-up land 700 to 1,000 m away, and the difference faded within about 500 m. Around the Arboretum and City Park it was about 1 °C. These differences are small and noisy, so read them as a hint rather than a measured cooling distance.

27.5 °C

Karura Forest canopy

28.8 °C

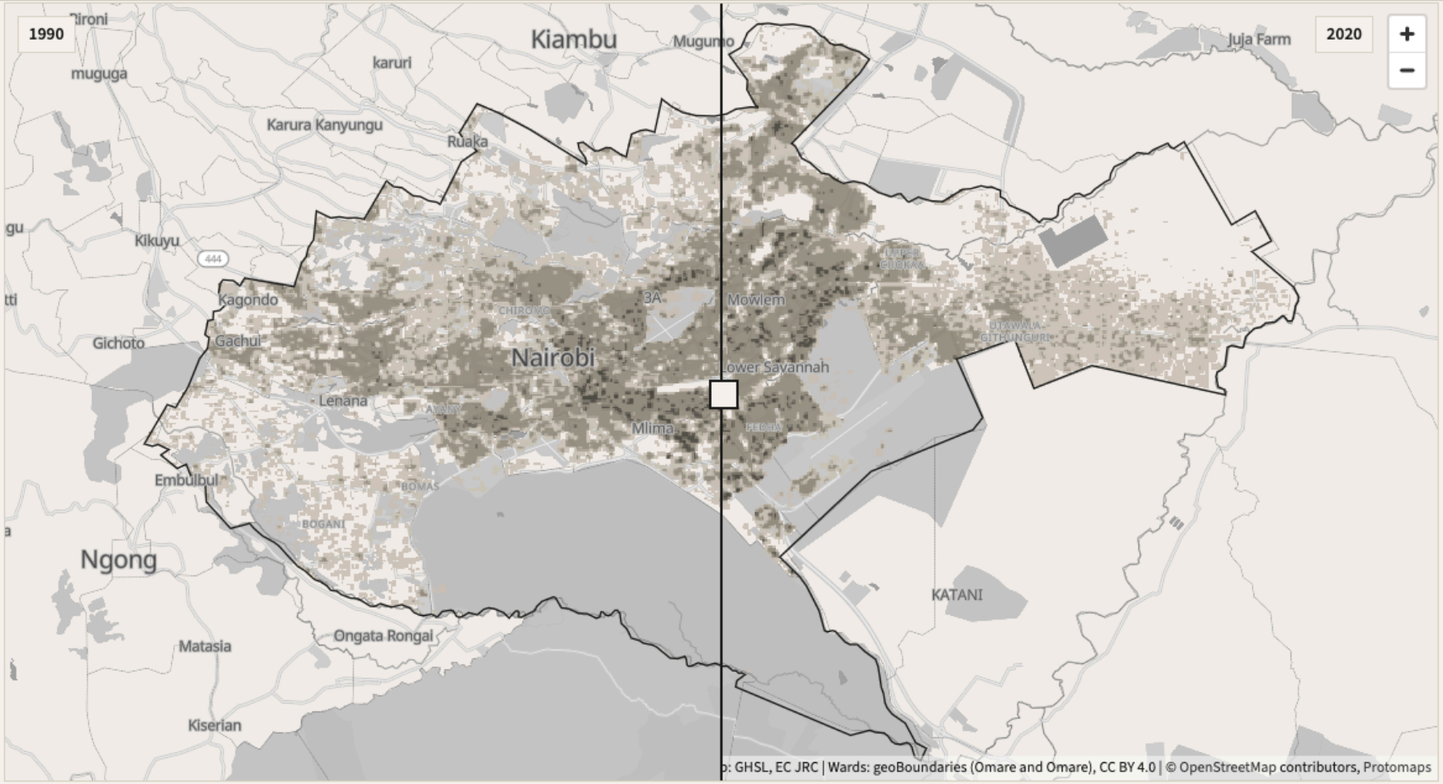
Nairobi Arboretum

2.1 °C

cooler: built-up land within 100 m of Karura against 700 to 1,000 m away

A small and noisy edge effect. Read it as a hint.

A growing city: built-up land rose from 66 km² to 101 km²



Built-up surface, GHSL 2020

5 % 25 % 50 %

No colour: under 5 %

Built-up surface, 1990 and 2020. Share of each 90 m cell covered by buildings and other built surfaces. Left of the handle: 1990. Right: 2020.

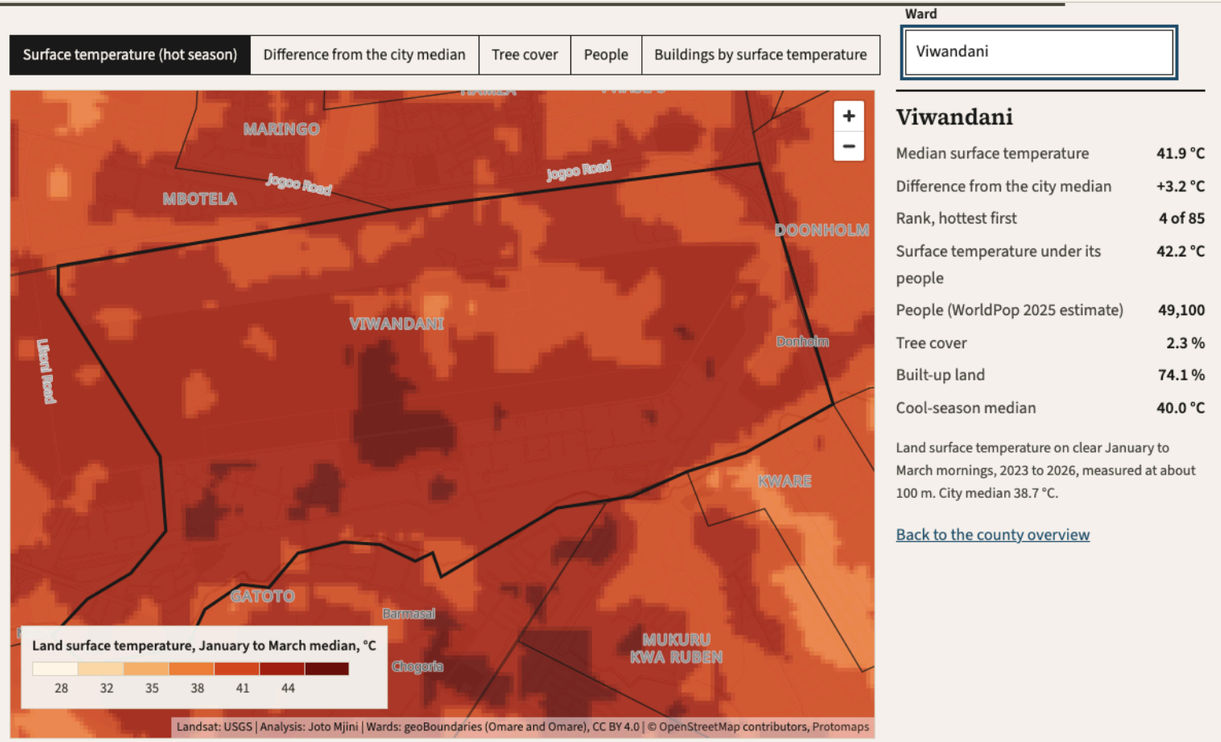
Source: Pesaresi and Politis (2023), GHS-BUILT-S R2023A, European Commission Joint Research Centre, CC BY 4.0. Basemap © OpenStreetMap contributors, Protomaps.

1.5 times

as much built-up land in 2020 as in 1990

The satellite record of surface temperature is too patchy to show a warming trend (Mann-Kendall p = 1).

Find your ward



Ward

Viwandani

Viwandani

Median surface temperature	41.9 °C
Difference from the city median	+3.2 °C
Rank, hottest first	4 of 85
Surface temperature under its people	42.2 °C
People (WorldPop 2025 estimate)	49,100
Tree cover	2.3 %
Built-up land	74.1 %
Cool-season median	40.0 °C

Land surface temperature on clear January to March mornings, 2023 to 2026, measured at about 100 m. City median 38.7 °C.

Back to the county overview

The ten wards where people live on the hottest ground

Land surface temperature weighted by where people live, January to March mornings 2023 to 2026

Rank	Ward	Surface temperature, people-weighted	People	Tree cover
1	Viwandani	42.2 °C	49,100	2 %
2	Kangemi	42.2 °C	11,700	13 %

All 85 wards: surface temperature, difference from the city median, rank, people, tree cover

Five layers, including buildings coloured by the surface temperature around them

Shareable link for any ward and layer

Next: rainfall, flooding and El Niño.

Data and credits

- Landsat 8 and 9 Collection 2 Level-2: U.S. Geological Survey, via Microsoft Planetary Computer
- Land cover: ESA WorldCover 2021 (Zanaga et al. 2022), CC BY 4.0
- Population: WorldPop 2025, R2025A (Bondarenko et al. 2025), CC BY 4.0
- Built-up area: GHS-BUILT-S R2023A, European Commission JRC, CC BY 4.0
- Buildings: Overture Maps Foundation; © OpenStreetMap contributors (ODbL)
- Wards: Omare and Omare via geoBoundaries, CC BY 4.0
- Air temperature: Open-Meteo, ERA5 (Copernicus, ECMWF), CC BY 4.0; El Niño index: NOAA CPC
- Basemap: © OpenStreetMap contributors, Protomaps

Story and code: [URL] | [repository link]